

Tanish Pal

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EDUCATION

Dr B R Ambedkar National Institute of Technology, Jalandhar

Bachelor of Technology in Electronics and Communications Engineering

Expected Graduation: Jun 2026

CGPA: 7.21/10

Rosemary Convent School, Bhatinda

CBSE – Intermediate

Apr 2021

Percentage: 79%

Army Public School, Bhatinda Cantt.

CBSE – Matriculate

Apr 2019

Percentage: 87.4%

SKILLS

Technical Skills: Cadence, Circuit Analysis, Control Systems, Embedded Systems (ESP32/Arduino), MATLAB, LTSpice, Verilog, IoT Integration, PCB Design

Soft Skills: Analytical Thinking, Problem Solving, Teamwork, Communication, Adaptability, Leadership

PROJECTS

Smart Waste Management System (IoT + Web App)

Jan 2025 – Apr 2025

- Developed an IoT-based smart dustbin using ESP32 with ultrasonic, gas, and MPU6050 sensors for real-time fill-level, odour, and tilt monitoring.
- Designed sensor calibration, filtering, and threshold logic to ensure reliable data acquisition under environmental noise and drift.
- Implemented efficient power management strategies for stable low-power embedded operation.
- Built a scalable Node.js + MongoDB backend and React.js dashboard with Google Maps integration for real-time geospatial tracking.
- Applied Dijkstra's algorithm to optimize waste collection routes, minimizing travel distance and operational redundancy.
- Established real-time communication using WebSockets and validated end-to-end data transmission reliability through prototype testing.
- Tech Stack: ESP32, Node.js, Express.js, MongoDB, React.js, Google Maps API, WebSockets

Reduction of Leakage Current in Domino Circuit | Cadence Virtuoso

Jan 2025

- Designed and simulated domino logic topologies in CMOS to analyze subthreshold and gate oxide leakage mechanisms.
- Implemented leakage reduction techniques including transistor sizing optimization, stacking effect utilization, and keeper modification.
- Performed DC and transient simulations to evaluate leakage-delay and power-noise trade-offs.
- Analyzed impact of supply voltage scaling on static power dissipation.
- Achieved measurable reduction in leakage power while maintaining noise margin and circuit stability.
- Tools: Cadence Virtuoso, LTSpice

High-Speed, Noise-Immune Domino Circuit Design & Simulation

Aug 2024 – Dec 2024

- Designed a 90nm CMOS domino logic circuit addressing speed-noise trade-off in wide fan-in dynamic logic architectures.
- Integrated delay-controlled keeper and Schmitt trigger to reduce contention current and enhance noise immunity.
- Improved propagation delay by 6.9% (26.3 ps → 24.5 ps) and increased Unity Noise Gain by 9.1%.
- Conducted PVT corner analysis and Monte Carlo simulations to ensure robustness across process and temperature variations.
- Benchmarked performance against conventional domino circuits to validate improvements in delay and noise tolerance.
- Tech Stack: Cadence Virtuoso, LTSpice, CMOS 90nm

EXTRA-CURRICULAR ACTIVITIES

- Finalist in the GDSC NITJ flagship hackathon; ranked among the top participants.
- Volunteered at Prayas NITJ, contributing to educational programs for underprivileged children.
- Coordinated teaching initiatives, organized events, and supported community welfare activities.
- Active member of NITJ Chess Hawks, contributing to tournament organization and promotion of strategic thinking.